🏆 Infosys SE Elite Crack Vault — Mock Test 01 (College ITM Master Full Simulation 01)

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Test Architecture: 54 Questions | 100 Minutes | Strict Sectional Cutoffs | Zero Negative Marking**
> **Sectional Breakdown:**
> - 🧠 **Section 1: Reasoning Ability** (15 Questions | 25 Mins | Cutoff: 70%+)
> - 📐 **Section 2: Mathematical Ability** (10 Questions | 35 Mins | Cutoff: 70%+)
> - 🔲 **Section 3: Verbal Ability** (20 Questions | 20 Mins | Cutoff: 65%+)
> - 💻 **Section 4: Pseudocode & Technical** (5 Questions | 10 Mins | Cutoff: 70%+)
> - 🧩 **Section 5: Puzzle Solving** (4 Questions | 10 Mins | Cutoff: 75%+)
> **Verification Score:** 100% Mathematically & Deductively Proven with Zero Hallucination

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> [!IMPORTANT]
> **Instructions for Candidate:**
> 1. You cannot navigate back to a completed section once its timer has expired.
> 2. There is no negative marking, so ensure every question is attempted.
> 3. Puzzles carry 2 to 3 marks each; allocate at least 2 minutes per puzzle.

Problem 01: Section 1: Reasoning Ability (Q01) 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q01` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Statements: All engineers are coders. Some coders are testers. No tester is a manager.
Conclusions: I. Some coders are not managers. II. Some engineers are testers. III. No engineer is a manager.
(a) Only I follows
(b) I and II follow
(c) I and III follow
(d) All follow

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Statement I: All engineers are coders, Some coders are testers, No tester is a manager. The subset of coders who are testers cannot be manag...

Problem 02: Section 1: Reasoning Ability (Q02) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q02 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Statements: Some pens are books. All books are chairs. No chair is a table.
Conclusions: I. Some pens are chairs. II. No book is a table. III. Some pens are not tables. IV. Some tables are pens.
(a) I and II only
(b) I, II and III only
(c) II, III and IV only
(d) All follow

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Statements: Some pens are books. All books are chairs. No chair is a table.
- Conclusion I: Some pens are chairs. Since some pens are books ...

Problem 03: Section 1: Reasoning Ability (Q03) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q03 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Data Sufficiency. For integers x and y, is x > y?
I. $x^2 > y^2$. II. x + y > 0.
(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are sufficient, neither alone
(d) Even both
together are not sufficient

Correct Answer

> [!IMPORTANT]

> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Question: Is x > y for integers x and y?
- Statement I: $x^2 > y^2$ gives |x| > |y|. If x = -3, y = 2, $x^2 = 9 > 4$ but x < y. If x = 3, y = 2, x ...

Problem 04: Section 1: Reasoning Ability (Q04) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q04 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Data Sufficiency. What is Riya's present age?
I. Riya is twice as old as her sister, who is 3 years younger than their brother.
II. Five years hence, the sum of the ages of Riya and her sister will be 40.
(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are sufficient, neither alone
(d) Even both
together are not sufficient

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Question: What is Riya's present age?
- Statement I: R = 2S, S = B - 3. Three unknowns, two equations. Insufficient.
- Statement II: (R + 5)...

Problem 05: Section 1: Reasoning Ability (Q05) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q05 | **Source:** College ITM Master Paper 1

Question Statement & Options:

In a certain code, MONDAY is written as NNOCBX. How is SUNDAY written in that code?
(a) TTOCBX
(b) TVOEBZ

(c) TTOCBZ
(d) RTOCBX

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Coding Pattern: M (+1) -> N, O (-1) -> N, N (+1) -> O, D (-1) -> C, A (+1) -> B, Y (-1) -> X. The alternating shift pattern is [+1, -1, +1, ...

Problem 06: Section 1: Reasoning Ability (Q06) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q06 | **Source:** College ITM Master Paper 1

Question Statement & Options:

In a code, "sun rises early" = "ka mo ti", "early bird catches" = "ti ra po", and "bird sun light" = "ra ka du".
What is the code for "rises bird light"?
(a) mo ra du
(b) mo po du
(c) ka ra du
(d) mo ti du

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Word-code comparison:
1) 'sun rises early' = 'ka mo ti'
2) 'early bird catches' = 'ti ra po' => Common word 'early' = 'ti'.
3) 'bird sun lig...

Problem 07: Section 1: Reasoning Ability (Q07) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q07 | **Source:** College ITM Master Paper 1

Question Statement & Options:

A solid cube painted red on all faces is cut into 64 identical small cubes (4 × 4 × 4). How many small cubes have exactly two faces painted?
(a) 12
(b) 24
(c) 36
(d) 8

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

For an n × n × n painted cube cut into unit cubes (here n = 4 since $4^3 = 64$):
Cubes with exactly two faces painted lie along the 12 edges (e...

Problem 08: Section 1: Reasoning Ability (Q08) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q08` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

In triangle ABC, the medians AD, BE and CF are drawn and meet at G. How many triangles are there in the figure in total?
(a) 12
(b) 14
(c) 16
(d) 18

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Standard median geometry counting:
In triangle ABC with medians AD, BE, CF intersecting at centroid G:
1) 6 small elementary triangles forme...

Problem 09: Section 1: Reasoning Ability (Q09) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q09 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Pointing to a man, Meera said, "His mother is the only daughter of my mother." How is Meera related to the man?
(a) Mother
(b) Sister
(c) Aunt
(d) Grandmother

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Meera said: 'His mother is the only daughter of my mother.'
The 'only daughter of my mother' when spoken by a woman (Meera) refers to Meera ...

Problem 10: Section 1: Reasoning Ability (Q10) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q10 | **Source:** College ITM Master Paper 1

Question Statement & Options:

If A + B means A is the mother of B, A − B means A is the brother of B, and A × B means A is the father of B, which expression shows that P is the maternal uncle of R?
(a) P + Q − R
(b) P − Q + R
(c) P × Q − R
(d) P − Q × R

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

P must be the maternal uncle of R (brother of R's mother).
- Q + R means Q is the mother of R.
- P - Q means P is the brother of Q.
Combinin...

Problem 11: Section 1: Reasoning Ability (Q11) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q11 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Ravi walks 10 m north, turns right and walks 6 m, turns right and walks 14 m, turns left and walks 3 m, then turns left and walks 4 m. Where is he now relative to the starting point?
(a) 9 m East
(b) 9 m West
(c) 11 m South-East
(d) 5 m North

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Step-by-step vector tracking from origin (0, 0):
1) Walks 10 m North -> (0, 10)
2) Turns right (East) and walks 6 m -> (6, 10)
3) Turns righ...

Problem 12: Section 1: Reasoning Ability (Q12) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q12 | **Source:** College ITM Master Paper 1

Question Statement & Options:

A person facing North-East turns 90° clockwise, then 135° anticlockwise, then 180° clockwise. Which direction is he facing now?
(a) North
(b) South

(c) South-West
(d) East

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Initial direction: North-East (45° clockwise from North).
1) +90° clockwise: 45° + 90° = 135° (South-East).
2) -135° anticlockwise: 135° - 1...

Problem 13: Section 1: Reasoning Ability (Q13) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q13` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Find the next term: 3, 7, 16, 35, 74, ?
(a) 143
(b) 153
(c) 149
(d) 158

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Series: 3, 7, 16, 35, 74, ?
Pattern: ×2 + k where k increments by 1 at each step:
- 3 × 2 + 1 = 7
- 7 × 2 + 2 = 16
- 16 × 2 + 3 = 35
- 35 × ...

Problem 14: Section 1: Reasoning Ability (Q14) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)

> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q14` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Find the next term: 4, 9, 25, 49, 121, 169, ?
(a) 289
(b) 259
(c) 361
(d) 323

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Series: 4, 9, 25, 49, 121, 169, ?
Notice the base numbers:
- $2^2 = 4$
- $3^2 = 9$
- $5^2 = 25$
- $7^2 = 49$
- $11^2 = 121$
- $13^2 = 169$

These are the squar...

Problem 15: Section 1: Reasoning Ability (Q15) 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q15` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Find the next letter: B, E, I, N, T, ?
(a) A
(b) Z
(c) B
(d) Y

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Letter positions in alphabet: B(2), E(5), I(9), N(14), T(20), ?
Differences:
- 5 - 2 = +3
- 9 - 5 = +4
- 14 - 9 = +5
- 20 - 14 = +6
Next dif...

SECTION 2: Technical Ability (Quantitative) (35 min)

Problem 16: Section 2: Mathematical Ability (Q16) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q16 | **Source:** College ITM Master Paper 1

Question Statement & Options:

In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?
(a) 5400
(b) 3600
(c) 10800
(d) 2700

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Word COMMITTEE contains 9 letters: C(1), O(1), M(2), I(1), T(2), E(2).
Vowels: O, I, E, E (4 vowels, with two E's).
Consonants: C, M, M, T, ...

Problem 17: Section 2: Mathematical Ability (Q17) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q17 | **Source:** College ITM Master Paper 1

Question Statement & Options:

A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?
(a) 90
(b) 100
(c) 105
(d) 120

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Committee of 5 from 5 boys and 4 girls with at least 2 boys and at least 2 girls.
Possible compositions (Boys, Girls):
- Case 1: 2 Boys and ...

Problem 18: Section 2: Mathematical Ability (Q18) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q18 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?
(a) 4/11
(b) 1/3
(c) 5/11
(d) 4/9

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Two fair dice thrown. Total outcomes with at least one 4:
Sample space S = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (1,4), (2,4), (3,4), (...

Problem 19: Section 2: Mathematical Ability (Q19) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)

> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q19` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?
(a) 3/11
(b) 1/4
(c) 2/11
(d) 5/22

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Total balls = 5 red + 4 blue + 3 green = 12 balls.
3 balls drawn at random without replacement.
Total ways = C(12, 3) = (12 × 11 × 10) / (3 ...

Problem 20: Section 2: Mathematical Ability (Q20) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q20` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:
(a) 7.1%
(b) 5.5%
(c) 8.6%
(d) 6.4%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Let Cost Price (CP) = 100.
Marked Price (MP) = 100 × 1.40 = 140.

Successive discounts of 10% and 15%:
Selling Price (SP) = 140 × (1 - 0.10) ...

Problem 21: Section 2: Mathematical Ability (Q21) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q21 | **Source:** College ITM Master Paper 1

Question Statement & Options:

A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, … (one person per day, starting with A). On which day is the job completed?
(a) 16th
(b) 17th
(c) 18th
(d) 19th

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Let total work = LCM(12, 18, 24) = 72 units.
Daily efficiencies:
- A = 72 / 12 = 6 units/day
- B = 72 / 18 = 4 units/day
- C = 72 / 24 = 3 u...

Problem 22: Section 2: Mathematical Ability (Q22) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q22 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:
(a) 90 km/h
(b) 108 km/h
(c) 120 km/h
(d) 72 km/h

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Total distance to clear = 240 m + 160 m = 400 m.
Let speed of faster train = v1 m/s and slower train = v2 m/s.
1) Running in opposite direct...

Problem 23: Section 2: Mathematical Ability (Q23) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q23 | **Source:** College ITM Master Paper 1

Question Statement & Options:

In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?
(a) 1520
(b) 1600
(c) 1360
(d) 1440

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Total registered voters = 20,000.
Voters who voted = 80% of 20,000 = 16,000.
Invalid votes = 5% of 16,000 = 800.
Valid votes = 16,000 - 800 ...

Problem 24: Section 2: Mathematical Ability (Q24) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q24 | **Source:** College ITM Master Paper 1

Question Statement & Options:

The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ₹310. Find the sum.
(a) ₹8,000
(b) ₹9,000
(c) ₹10,000
(d) ₹12,000

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Difference between CI and SI for 3 years at rate r%:
Formula: Difference = $P \times (r / 100)^2 \times (3 + r / 100)$.
Given r = 10% and Difference = ₹3...

Problem 25: Section 2: Mathematical Ability (Q25) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q25` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

₹12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ₹2,270. How much was invested at simple interest?
(a) ₹4,000
(b) ₹5,000
(c) ₹6,000
(d) ₹7,000
Read the passage and answer Q26–30.
In software organisations, "technical debt" is often treated as a purely engineering concern, a messy residue of hurried decisions that developers will eventually tidy up. This framing is misleading. Debt, in the financial sense, is not inherently harmful; a company that borrows to seize a market opportunity may be acting shrewdly. The problem arises when the interest, meaning the extra effort every future change demands, is never acknowledged on the balance sheet. Teams then find that features which once took days now take weeks, yet managers, seeing only the delivered functionality, attribute the slowdown to declining developer productivity. Proponents of disciplined refactoring argue that a fixed share of every sprint should be reserved for repayment. Critics counter that such quotas are arbitrary and encourage cosmetic rewrites that leave real risks untouched. A more defensible approach, some suggest, is to make debt visible by tracking how often particular modules cause delays or defects, and to repay only where the interest is demonstrably high.

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Let sum invested at 8% SI = ₹x.
Then sum invested at 10% CI = ₹(12,000 - x).
Time = 2 years.
- Simple Interest at 8% for 2 years = (x × 8 × ...

SECTION 3: Verbal Ability (20 min)

Reading Comprehension Passage (Q26–Q30):

```
Read the passage and answer Q26–30.
In software organisations, "technical debt" is often treated as a purely engineering concern, a messy
residue of hurried decisions that developers will eventually tidy up. This framing is misleading. Deb
the financial sense, is not inherently harmful; a company that borrows to seize a market opportunity
be acting shrewdly. The problem arises when the interest, meaning the extra effort every future chang
demands, is never acknowledged on the balance sheet. Teams then find that features which once took
days now take weeks, yet managers, seeing only the delivered functionality, attribute the slowdown to
declining developer productivity. Proponents of disciplined refactoring argue that a fixed share of e
sprint should be reserved for repayment. Critics counter that such quotas are arbitrary and encourage
cosmetic rewrites that leave real risks untouched. A more defensible approach, some suggest, is to ma
debt visible by tracking how often particular modules cause delays or defects, and to repay only wher
the interest is demonstrably high.
```

Problem 26: Section 3: Verbal Ability (Q26) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q26` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

The central idea of the passage is that:
(a) Technical debt is harmful and must be eliminated at once
(b) Technical debt is not necessarily bad, but its recurring costs must be made visible and repaid selectively
(c) Managers, not developers, are responsible for technical debt
(d) Fixed refactoring quotas are the best way to manage debt

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

The passage argues that technical debt is analogous to financial debt: not inherently evil, but dangerous when its recurring carrying costs ...

Problem 27: Section 3: Verbal Ability (Q27) 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q27 | **Source:** College ITM Master Paper 1

Question Statement & Options:

According to the passage, managers blame developer productivity because they:
(a) do not trust developers
(b) see only the delivered features and not the hidden extra effort
(c) are unaware that software can be refactored
(d) have reserved too little time for sprints

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

The passage states explicitly: 'yet managers, seeing only the delivered functionality, attribute the slowdown to declining developer product...

Problem 28: Section 3: Verbal Ability (Q28) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q28 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Which statement would critics of fixed repayment quotas most likely agree with?
(a) Repayment effort should target genuine risks rather than meet a set proportion
(b) Refactoring should be abandoned entirely
(c) Cosmetic rewrites are the most valuable form of repayment
(d) Sprints should be lengthened

Correct Answer

> [!IMPORTANT]

> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Critics argue that fixed quotas 'are arbitrary and encourage cosmetic rewrites that leave real risks untouched.' Hence, they would advocate ...

Problem 29: Section 3: Verbal Ability (Q29) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q29 | **Source:** College ITM Master Paper 1

Question Statement & Options:

In the passage, "interest" most nearly refers to:
(a) money paid to lenders
(b) the recurring additional effort imposed on future changes
(c) the curiosity of developers
(d) the profit from a market opportunity

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

The passage defines interest directly: 'the interest, meaning the extra effort every future change demands'....

Problem 30: Section 3: Verbal Ability (Q30) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q30 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Which of the following is NOT supported by the passage?
(a) Borrowing can sometimes be a shrewd decision
(b) A slowdown in delivery always indicates falling developer productivity
(c) Tracking defect-prone modules can help identify costly debt
(d) Debt is often left unrecorded in practice
Sentence Correction (Q31–35). Choose the best replacement for the bold part.

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

The passage explicitly notes that a delivery slowdown is often caused by accumulated technical debt rather than declining developer producti...

Problem 31: Section 3: Verbal Ability (Q31) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q31` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Neither the manager nor the engineers was aware of the revised deadline.
(a) No improvement
(b) engineers were
(c) engineers has been
(d) engineer were

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

With correlative conjunctions 'neither... nor', the verb agrees in number with the nearer subject. Here, the nearer subject is 'engineers' (...

Problem 32: Section 3: Verbal Ability (Q32) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q32` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Had the team would have tested the patch thoroughly, the outage could have been avoided.
(a) No improvement
(b) Had the team tested

(c) Has the team tested
(d) If the team would test

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Inverted third conditional structure uses 'Had + subject + past participle'. The conditional clause cannot contain 'would have'. Correct: 'H...

Problem 33: Section 3: Verbal Ability (Q33) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q33` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Hardly the server had restarted when the alerts began flooding in.
(a) No improvement
(b) had the server restarted when
(c) did the server restarted when
(d) the server
restarted than

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Negative adverbial 'Hardly' placed at the beginning of a clause triggers negative inversion: 'Hardly had + subject + past participle... when...

Problem 34: Section 3: Verbal Ability (Q34) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q34` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

The number of applicants are increasing, but a large number of them lacks the required skills.
(a) are ... lacks
(b) is ... lack
(c) is ... lacks
(d) are ... lack

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

'The number of' denotes a singular collective quantity and takes a singular verb ('is increasing'). In contrast, 'a large number of' denotes...

Problem 35: Section 3: Verbal Ability (Q35) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q35` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Each of the interns, along with their mentor, were invited to present.
(a) No improvement
(b) was
(c) are
(d) have been
Para-jumbles (Q36–38). Arrange the sentences in the correct order.

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

The true grammatical subject of the sentence is 'Each' (distributive pronoun, singular). Parenthetical prepositional phrases like 'along wit...

Problem 36: Section 3: Verbal Ability (Q36) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)

> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q36` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

P: However, the prototype failed under real-world load.
Q: The engineers were confident that their design was flawless.
R: Post-mortem analysis revealed that no load testing had been performed.
S: Consequently, the launch was postponed by three months.
(a) QPRS
(b) QRPS
(c) PQRS
(d) QSPR

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Coherent narrative flow: Q introduces initial state ('The engineers were confident...'), P introduces the contrast ('However, the prototype ...

Problem 37: Section 3: Verbal Ability (Q37) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q37` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

P: Rather, they encourage people to think about how the story might have unfolded differently.
Q: Counterfactual questions such as "What if the treaty had never been signed?" are popular with historians.
R: They do not aim to predict what would actually have happened.
S: Such speculation, when disciplined, can illuminate the forces that shaped real events.
(a) QRPS
(b) QPRS
(c) RQPS
(d) QRSP

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Logical progression: Q introduces the subject ('Counterfactual questions... are popular with historians'), R defines their negative boundary...

Problem 38: Section 3: Verbal Ability (Q38) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q38 | **Source:** College ITM Master Paper 1

Question Statement & Options:

P: The first industrial robots were deployed on car assembly lines in the 1960s.
Q: Today, collaborative robots work safely alongside humans.
R: These early machines were caged off, since they could not sense human presence.
S: Advances in sensors and software have since removed this limitation.
(a) PRSQ
(b) PQRS
(c) PSRQ
(d) RPSQ
Vocabulary, Synonyms and Antonyms (Q39–45).

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

Chronological progression: P establishes historical origin in the 1960s ('The first industrial robots were deployed...'), R describes their ...

Problem 39: Section 3: Verbal Ability (Q39) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q39 | **Source:** College ITM Master Paper 1

Question Statement & Options:

The auditor's meticulous approach uncovered discrepancies others had missed. "Meticulous" means:
(a) hasty
(b) painstakingly careful
(c) aggressive
(d) casual

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

'Meticulous' means showing great attention to detail; very careful and precise. Hence, 'painstakingly careful'....

Problem 40: Section 3: Verbal Ability (Q40) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q40` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

The contract was deliberately worded to obfuscate the penalty clauses. "Obfuscate" means:
(a) highlight
(b) simplify
(c) make obscure
(d) cancel

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

'Obfuscate' means to make obscure, unclear, or unintelligible....

Problem 41: Section 3: Verbal Ability (Q41) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q41` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Fill in the blank: The startup's success proved _____; within a year it had vanished.
(a) enduring
(b) ephemeral
(c) robust
(d) inevitable

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

'Ephemeral' means lasting for a very short time; transitory. Since the startup vanished within a year, its success was ephemeral....

Problem 42: Section 3: Verbal Ability (Q42) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q42` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Synonym of UBIQUITOUS:
(a) rare
(b) omnipresent
(c) ancient
(d) hidden

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

'Ubiquitous' means present, appearing, or found everywhere; omnipresent....

Problem 43: Section 3: Verbal Ability (Q43) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q43` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Synonym of RECALCITRANT:
(a) obedient
(b) obstinate
(c) cheerful
(d) timid

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

'Recalcitrant' means having an obstinately uncooperative attitude towards authority or discipline; obstinate....

Problem 44: Section 3: Verbal Ability (Q44) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q44` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Antonym of LOQUACIOUS:
(a) talkative
(b) taciturn
(c) eloquent
(d) boisterous

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

'Loquacious' means tending to talk a great deal; talkative. The antonym is 'taciturn' (reserved or uncommunicative in speech)....

Problem 45: Section 3: Verbal Ability (Q45) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q45` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Antonym of MAGNANIMOUS:
(a) generous
(b) noble
(c) petty
(d) tolerant

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

'Magnanimous' means generous or forgiving, especially towards a rival or someone less powerful. The antonym is 'petty' (narrow-minded, spite...

SECTION 4: Pseudo Code (10 min)

Problem 46: Section 4: Pseudocode & Technical (Q46) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q46 | **Source:** College ITM Master Paper 1

Question Statement & Options:

What is printed?
int f(int n) {
if (n <= 1) return n;
return f(n-1) + 2 * f(n-2);
}
print(f(6))
(a) 15
(b) 21
(c) 27
(d) 31

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Recursive function trace:
- f(0) = 0
- f(1) = 1
- f(2) = f(1) + 2·f(0) = 1 + 0 = 1
- f(3) = f(2) + 2·f(1) = 1 + 2(1) = 3
- f(4) = f(3) + 2·f...

Problem 47: Section 4: Pseudocode & Technical (Q47) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q47 | **Source:** College ITM Master Paper 1

Question Statement & Options:

What is printed?
x = 45; count = 0;
while (x != 0) {
x = x & (x - 1);
count++;
}
print(count ^ (45 >> 2))
(a) 4
(b) 11
(c) 15
(d) 7

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Brian Kernighan's bit-counting loop clears the lowest set bit in each iteration.
For x = 45 = 32 + 8 + 4 + 1 = 101101_2 (4 set bits):
Loop ex...

Problem 48: Section 4: Pseudocode & Technical (Q48) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q48 | **Source:** College ITM Master Paper 1

Question Statement & Options:

What is printed?
s = 0
for i = 1 to 5
for j = i to 5
if ((i + j) % 3 == 0) continue
if (j == 5) break
s = s + i * j

end for
end for
print(s)
(a) 38
(b) 42
(c) 46
(d) 50

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Nested loops simulation:
- i=1: j=1 (sum=2, add 1); j=2 ((1+2)%3==0, continue); j=3 (sum=4, add 3); j=4 (sum=5, add 4); j=5 ((1+5)%3==0, con...

Problem 49: Section 4: Pseudocode & Technical (Q49) 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q49 | **Source:** College ITM Master Paper 1

Question Statement & Options:

A stack is used (operands are evaluated left to right). What is printed?
push(3); push(5); push(7);
x = pop();
push(x * 2);
push(pop() + pop());
print(pop() - pop())
(a) 14
(b) 16
(c) 19
(d) 22

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Stack execution trace (LIFO):
1) push(3) -> [3]

2) push(5) -> [3, 5]
3) push(7) -> [3, 5, 7]
4) x = pop() => x = 7; stack = [3, 5]
5) push(7...

Problem 50: Section 4: Pseudocode & Technical (Q50) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q50 | **Source:** College ITM Master Paper 1

Question Statement & Options:

What is the final content of arr? (indexing starts at 0; updates are in place)
arr = [4, 1, 3, 2]
for i = 0 to 3
arr[i] = arr[i] + arr[arr[i] % 4]
end for
(a) [8, 2, 5, 7]
(b) [8, 2, 5, 6]
(c) [8, 5, 5, 7]
(d) [5, 2, 5, 7]

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((A))**

Step-by-Step Mathematical & Logical Derivation

In-place array update with arr = [4, 1, 3, 2]:
- i = 0: arr[0] = arr[0] + arr[arr[0] % 4] = 4 + arr[4 % 4] = 4 + arr[0] = 4 + 4 = 8. arr = [...

SECTION 5: Numeric Puzzle Solving (10 min)

Problem 51: Section 5: Puzzle Solving (Q51) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** MOCK01_Q51 | **Source:** College ITM Master Paper 1

Question Statement & Options:

Matchstick. A 2 × 2 grid of squares is made using 12 matchsticks (giving 4 small squares and 1 large square). What is the minimum number of sticks to remove so that no square of any size remains?
(a) 2

(b) 3
(c) 4
(d) 5

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

In a 2 × 2 matchstick grid (12 sticks total, forming four 1×1 squares and one 2×2 square):
To eliminate all 5 squares with the minimum numbe...

Problem 52: Section 5: Puzzle Solving (Q52) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q52` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Circular wheel. Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.
Wheel 1: T = 3, R = 5, B = 4, L = 7, Centre = 14
Wheel 2: T = 2, R = 6, B = 5, L = 3, Centre = 7
Wheel 3: T = 4, R = 3, B = 6, L = 9, Centre = ?
(a) 21
(b) 27
(c) 30
(d) 33

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((C))**

Step-by-Step Mathematical & Logical Derivation

Determining the circular wheel formula:
- Wheel 1: T=3, R=5, B=4, L=7, Centre=14.
Formula: Centre = (T × B) + L - R = (3 × 4) + 7 - 5 = 12...

Problem 53: Section 5: Puzzle Solving (Q53) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)

> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q53` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Matrix logic. Find the missing number.
3
4
25
5
12
169
6
8
?
(a) 64
(b) 100
(c) 196
(d) 144

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Matrix logic pattern:
Row 1: 3, 4, 25 => $3^2 + 4^2 = 9 + 16 = 25$.
Row 2: 5, 12, 169 => $5^2 + 12^2 = 25 + 144 = 169$.
Row 3: 6, 8, ? => $6^2 + 8^2 = ...$

Problem 54: Section 5: Puzzle Solving (Q54) 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Full-Length Timed Simulation)
> **Question ID:** `MOCK01_Q54` | **Source:** `College ITM Master Paper 1`

Question Statement & Options:

Pattern. Find the next term: 2, 12, 36, 80, 150, ?
(a) 216
(b) 252
(c) 240
(d) 294

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option **((B))**

Step-by-Step Mathematical & Logical Derivation

Sequence: 2, 12, 36, 80, 150, ?
Pattern: $n^3 + n^2$ for n = 1, 2, 3, 4, 5, ...
- n = 1: $1^3 + 1^2 = 1 + 1 = 2$
- n = 2: $2^3 + 2^2 = 8 + 4 = 12$
- n =...

SECTION 6: English Grammar (10 min)
